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B.Tech. DEGREE EXAMINATION, APRIL 2015.

Third Semester

Computer Science and Engineering

OBJECT ORIENTED PROGRAMMING AND DESIGN

(2013-2014 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Outline the disadvantages of procedural programming.
2. Explain Lvalue and Rvalues.
3. State the usage of Constructor and Destructor.
4. Explain abstract class.
5. Write the function for malloc() and free().
6. Write sample C++ code to display the length of string.

7. Outline the different types of file modes.
8. List the type of errors encountered while reading (Or) writing file.

9. Explain swim lane and life lane in UML.
10. State the usage of adapter pattern design.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each unit.

All questions carry equal marks.

UNIT I

11. Illustrate various control structures in C++.

Or

12. Discuss various building blocks for object oriented programming language.

UNIT II

13. Illustrate binary operator overloading function with a sample example.

Or

14. Explain multiple inheritances with a sample example.

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UNIT III

15. Discuss different types of array declarations with a sample example.

Or

16. Design a program to print the output of various string handling methods.

UNIT IV

17. Develop a program to read and write array of integers into binary file.

Or

18. Discuss exception handling in C++.

UNIT V

19. Construct the UML usecase diagram to withdraw amount in ATM banking.

Or

20. Design Sequence and Collaboration diagram for issue and return of books in library.

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UNIT IV

17. (a) Explain in detail about exception handling in C++ using sample code. (6)

(b) Write a C++ program using function template to find the product of two integer or floating point type of data. (5)

Or

18. Create a class Player with data members are playerid, playername, playergame and member functions are getplayer(), Searchplayer() and displayplayer(). Use file to store N employers information and perform those operation.

UNIT V

19. Derive the use case model for student course registration system in a University with proper interaction and activity diagram which include the following tasks Publication of Courses offered, student enrollment for courses, Attendance maintenance, Conduct of Exams and marks maintenance and Result publication.

Or

20. (a) Explain in detail about patterns with an example. (4)

(b) How an activity and state chart diagrams are used for modeling? (7)

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B.Tech. DEGREE EXAMINATION, NOVEMBER 2015.

Third Semester

Computer Science and Engineering

OBJECT ORIENTED PROGRAMMING AND DESIGN

(2013-2014 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Define Encapsulation and Data hiding.
2. State the advantages of inline functions in C++?
3. What is an abstract class?
4. What are the difference between default and parameterized constructors?
5. What is pure virtual function? Give an example.
6. How is a string object created?
7. What is Uncaught Exception? What is the causes of it?
8. What are the three standard template library adapters?

9. What do you know about use case model?
10. Write any two goodness criteria of object oriented design.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE question from each unit.

All questions carry equal marks.

UNIT I

11. (a) Explain the basic concepts of Object oriented programming. (7)
- (b) How are pointers used to access the class members? Explain with an example. (4)

Or

12. (a) What are manipulators? Explain in detail about the various manipulators used for I/O operations. (7)
- (b) Explain in detail about friend function in C++ with an example. (4)

UNIT II

13. Create a class Vector to hold an array of integers and perform the following operations using constructor for dynamic allocation of array based on size given as argument
- (a) Create two vectors using constructor

- (b) Add the vectors and return the resultant vector
- (c) Multiply all elements in the vector by a given number X and display the resultant vector.

Or

14. Write a C++ program to find the area of various 2D shapes such as square, rectangle, triangle, circle and ellipse using function overloading

UNIT III

15. (a) Explain in detail about the various run time casting in C++. (6)
- (b) Write a C++ program for calculating the area of rectangle and circle using run time polymorphism. (5)

Or

16. (a) Write a program in C++ to read two strings and perform the following string manipulation function (7)

- (i) Find the long string
- (ii) Compare the two strings
- (iii) Concatenate them.
- (b) Explain in detail about dynamic objects. How are they created? (4)

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B.Tech. DEGREE EXAMINATION,
NOVEMBER/DECEMBER 2014.

Third Semester

Computer Science and Engineering

OBJECT ORIENTED PROGRAMMING AND DESIGN

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. State any four advantages of object oriental programming.
2. Write the properties of friend function.
3. What is destructor?
4. What is function overloading? Give example.
5. Explain how to create dynamic objects.
6. Write a small program explaining Virtual functions.

7. What are the various modes of file operation?
8. What is a class template? How to define it?
9. What are the advantages of UML?
10. What are the Patterns?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Explain the major principles of object oriented programming with illustrations and neat sketches.

Or

12. Explain the decision making statements of C++.

UNIT II

13. Explain binary operator overloading with example program.

Or

14. Explain any two types of inheritance with example program.

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UNIT III

15. What is the use of new and delete operators Explain briefly overloading of the new and delete operator?

Or

16. Explain Virtual function with a program.

UNIT IV

17. Explain the file pointers and file manipulators.

Or

18. Explain Function Template with good example program.

UNIT V

19. Explain the interaction and state chart diagram c UML.

Or

20. Explain the concept of Interaction Modeling.

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UNIT IV

17. Explain the steps of file operation with example.

Or

18. Write a program to create template – based stack. Store float and integer in it.

UNIT V

19. Draw and explain the class diagram for a simple banking application.

Or

20. Explain interaction diagrams with an example.

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B.Tech. DEGREE EXAMINATION, APRIL/MAY 2016.

Third Semester

Computer Science and Engineering

OBJECT ORIENTED PROGRAMMING AND DESIGN

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Write few concepts of object – oriented programming.
2. What is function overloading?
3. What is meant by copy constructor?
4. Describe the syntax of multiple inheritance. When do we use such an inheritance?
5. Give an example to create array of objects using pointers.

6. What is the need of virtual function?
7. What is command line argument? Give an example.
8. What are the advantages of using exceptions?
9. What is meant by activity diagram?
10. Define interaction modelling.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE question from each Unit.

UNIT I

11. Write a program to evaluate the following function

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

Or

12. (a) What is a friend function. What are the merits and demerits of using friend function? (5)

- (b) Write a program to explain the inline function. (6)

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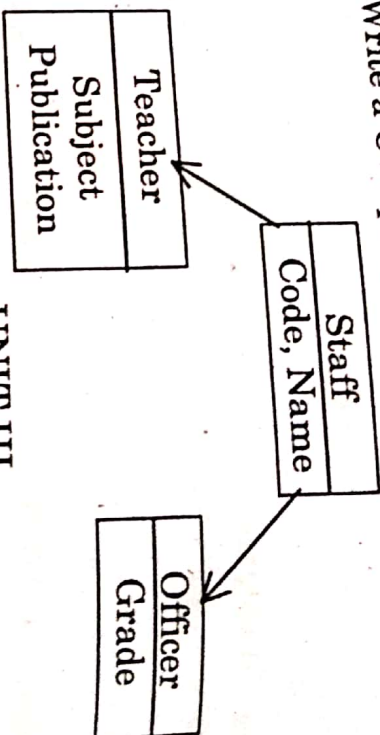
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UNIT II

13. Write a program to add two complex numbers using binary operator overloading concept.

Or

14. Write a C++ program to implement the following (17)



UNIT III

15. (a) Write a program to initialize the private numbers and display them without the use of member function. (5)

- (b) Write a program to declare void pointers and access member variables using void pointers (6)

Or

16. (a) Write a program to create dynamic object (5)
- (b) Write a program to invoke function using scope access operator. (6)

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**B.Tech. DEGREE EXAMINATION,
NOVEMBER/DECEMBER 2016.**

Third Semester

Computer Science and Engineering

OBJECT ORIENTED PROGRAMMING AND DESIGN

(2013 – 2014 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. What is data hiding?
2. What is a friend function?
3. Define virtual class.
4. What is an abstract class?
5. What is meant by an array? Give an example.
6. State the use of new and delete operators.

7. Mention the use of templates.
8. What is a file mode?
9. What is UML?
10. State the use of class diagram.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, choosing ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Explain the structure of the C++ program with an example.

Or

12. What is meant by function overloading? Explain in detail.

UNIT II

13. Discuss about dynamic initialization using constructors in detail.

Or

14. What is inheritance? Explain in detail.

UNIT III

15. Discuss about pointer to derived classes and base classes.

Or

16. Explain about declaring and initializing string objects in detail.

UNIT IV

17. Discuss command line arguments in detail.

Or

18. Explain about Exception handling in detail.

UNIT V

19. Discuss about interaction diagrams in detail.

Or

20. Explain about state chart diagrams in detail.

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B.Tech. DEGREE EXAMINATION, APRIL/MAY 2017.

Third Semester

Computer Science and Engineering

OBJECT ORIENTED PROGRAMMING AND DESIGN

(2013-14 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Compare structured programming and object oriented programming.
2. Mention the use of array of objects.
3. Define copy constructors.
4. What is meant by virtual class?
5. How can we access private variables using pointers?
6. Why do we need dynamic objects?

7. What is the need of templates?
8. List various file operation modes.
9. Where is activity diagrams used?
10. What are the various types of interaction diagrams?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Write a C++ program to find matrix inverse using functions.

Or

12. Write C++ programs that use both recursive and non-recursive functions.

- (a) Factorial of a given number
- (b) GCD of two given integers.

UNIT II

13. Write a C++ program to overload unary minus operator.

Or

14. Write a C++ program that illustrates hybrid inheritance.

UNIT III

15. Write a C++ program to find maximum among five numbers using array of pointers.

Or

16. Write a C++ program that implements area of rectangle and triangle using abstract classes.

UNIT IV

17. Write a C++ program that illustrates class templates.

Or

18. Explain exception handling in detail.

UNIT V

19. Explain various UML diagrams in detail.

Or

20. Describe various methodology used for OOAD

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B.Tech. DEGREE EXAMINATION, NOVEMBER 2017.

Third Semester

Computer Science and Engineering

OBJECT ORIENTED PROGRAMMING AND DESIGN

(2013 – 14 Onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. What are the advantages of OOP?
2. What is the use of the word friend keyword in C++?
3. Write some special characteristics of constructor.
4. List out the operators that cannot be overloaded.
5. List the two types of polymorphism? Give an example for each.
6. How are virtual functions declared in C++?
7. What are the three keywords used for exception handling mechanism?

8. Explain the usage of template in C++.
9. What do you know about UML?
10. What is interaction modelling?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Explain the features of object oriented programming. Describe how each of these is implemented in C++.

Or

12. Explain how to overload a function with an example program in C++.

UNIT II

13. Explain the different types of Constructors that are available in C++.

Or

14. What is the significance of inheritance and explain the different types of inheritance with a programming example for each type.

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UNIT III

15. Explain the use of Virtual Functions with an example program in C++.

Or

16. Explain how to concatenate two strings using an example program in C++.

UNIT IV

17. Explain in detail about exception handling mechanism and write a program in C++ illustrate the divide by zero exception.

Or

18. Explain the file handling techniques in C++.

UNIT V

19. Describe in detail: Object oriented analysis design methodology.

Or

20. Explain how to use the activity diagrams interaction diagrams.

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UNIT V

19. (a) Explain in detail about class diagrams. (6)
(b) Explain interaction modelling with example. (5)

Or

20. Explain in detail about object oriented system design and methodology. (11)

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B.Tech. DEGREE EXAMINATION, MAY 2018.

Third Semester

Computer Science and Engineering

OBJECT ORIENTED PROGRAMMING AND DESIGN

(2013-2014 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL the questions.

All questions carry equal marks.

1. List any four features of object oriented programming?
2. Define data abstraction.
3. What is the need of destructor?
4. What is the use of operator overloading?
5. What is abstract class?
6. What is friend function?
7. List the advantages of generic programming.

8. What is 'throw'? What is its use?
9. What are the main activities in design process?
10. How do you present UML operation?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. (a) Explain the principles of object oriented programming. (5)
- (b) Explain function call by-reference method with example. (6)

Or

12. (a) Explain the usage of Bit fields and manipulators. (6)
- (b) Explain function overloading with example. (5)

UNIT II

13. (a) Explain the use of virtual base class in inheritance. (6)
- (b) Write short notes on copy constructors. (5)

Or

14. (a) Explain operator over loading with example. (6)
- (b) Write short notes on common constructor. (5)

UNIT III

15. (a) Explain in detail about pointer to base and derived classes. (6)
- (b) Explain dynamic objects with a suitable example. (5)

Or

16. (a) Explain the procedure for accessing private members of an object with pointers. (6)
- (b) Explain the purpose of including virtual functions in derive classes. (5)

UNIT IV

17. (a) Explain the random access file operations. (6)
- (b) Explain the function templates with example. (5)

Or

18. (a) Explain the fundamentals of exception handling. (6)
- (b) Write short notes on file pointers and manipulators. (5)

UNIT V

19. (a) Draw and explain Class diagram for a library system.
(b) Draw and explain Sequence diagram for issue of book.

Or

20. Explain :
(a) State chart diagram
(b) Interaction modeling
(c) - OOAD methodology.

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B.Tech. DEGREE EXAMINATION, NOVEMBER 2018.

Third Semester

Computer Science and Engineering

OBJECT ORIENTED PROGRAMMING AND DESIGN

(2013-2014 onwards batches)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Describe extraction and insertion operator.
2. Define L Value and R Value.
3. What is copy constructor?
4. Describe virtual base class with example.
5. How to set pointer to class and object?.
6. What is binding? Name the types in binding.

UNIT II

7. Write about error handling functions.
8. What is generic function?
9. How to draw an use case model?
10. Mention the differences between Sequence diagram and Activity diagram.

PART B.—(5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Explain the following :
 - (a) Key concepts of object oriented programming.
 - (b) Usage of OOP.

Or

12. (a) Explain in OOPs how objects can be passed as function arguments with supporting program.
- (b) Explain the concepts of friend functions with suitable program.

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13. Explain overloading of one unary and any two binary operators with examples.

Or

14. Define Inheritance and its types with suitable examples. How do we resolve ambiguity in multiple inheritance?

UNIT III

15. Explain the use of new and delete operator with suitable example.

Or

16. Explain virtual functions. Illustrate the use of base pointer to point to different objects for executing different versions of the virtual functions.

UNIT IV

17. What is Template? Mention the need of template. Explain overloading of function templates.

Or

18. Explain Exception handling in C++ with various constructs supported by it. How to control uncaught exceptions?

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UNIT IV

17. Write an object oriented program in C++ using a class template to read any five parameterized data type such as float and integer, and print the average.

Or

18. Write a program that merges lines alternately from two files and writes the results to new file. If one file has less number of lines than the other, the remaining lines from the larger file should be simply copied into the target file.

UNIT V

19. Describe in detail: Object oriented analysis and design methodology.

Or

20. Explain how to use state chart diagram and class diagrams.

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B. Tech. DEGREE EXAMINATION, APRIL/MAY 2019.

Third Semester

Computer Science and Engineering

OBJECT ORIENTED PROGRAMMING AND DESIGN

(2013-14 onwards Regulation)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Differentiate Procedural programming and OO programming.
2. How to write an inline function?
3. What is destructor?
4. List out the operators that cannot be overloaded.
5. What is dynamic binding?

6. Compose pure virtual functions? Where are they used?
7. What are two types of file pointers used in C++?
8. What are the file stream classes in C++?
9. What do you know about activity diagram?
10. What do you know about OOD goodness criteria?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Explain the key concepts of object oriented programming. Describe how each of these is implemented in C++.

Or

12. Explain the various data types/ control statements/ operators/ storage classes that are available in C++ with neat illustration of each.

UNIT II

13. Explain the single inheritance, multiple inheritance and multilevel inheritances in C++ with an example.

Or

14. Develop a function that computes and displays the area of a rectangle and the area of a square. Declare this function as a friend function in Rectangle and square classes and compute the area of a rectangle and square objects from the main function.

UNIT III

15. Explain how to access private members with pointers in C++ with an example.

Or

16. Write a program to overload "+" operator to perform string concatenation operation and polynomial addition operation.

UNIT V

19. Discuss UML Class diagram in detail. (11)

Or

20. Explicate interaction modelling in UML with a suitable example. (11)

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B.Tech. DEGREE EXAMINATION, NOVEMBER 2019.

Third Semester

Computer Science and Engineering

Elective – OBJECT ORIENTED PROGRAMMING AND DESIGN

(2013-14 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL the questions.

All questions carry equal marks.

1. List the characteristics of procedure - oriented programming.
2. Differentiate between classes and objects.
3. Define the term default constructor.
4. What operators can and cannot be overloaded?
5. Comprehend void pointers. State the advantages of void pointers.

6. Define the term pure virtual function.
7. Annotate streams.
8. What are the tasks to be handled by exception handling?
9. Define use case model. List the elements of use case model.
10. Mention the purpose of using statechart diagram.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE question from each Unit.

All questions carry equal marks.

UNIT I

11. Annotate the basic concepts of object oriented programming. (11)
- Or
12. State the special characteristics of static data members and static member functions. Develop a class employee with two data members (eid and name) and two member functions (read() to read new data and write() to display the data). Write a program to find how many times the member functions are called by using static data member and display the count. (11)

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UNIT II

13. Define constructors. Create a class copycontent with two integer members, a parameterized constructor to initialize the data members, a copy constructor and a print () function to display the data members. Declare an object 'S' with values 100 and 200, create a new object 'D' and initialize using value of object 'S' by copy constructor. (11)

Or

14. Write a C++ program to overload + and - binary operators to do arithmetic operations between two objects. (11)

UNIT III

15. Explore the purpose of array of objects with an example. (11)

Or

16. Design a C++ program to define three classes A, B and C. The class C is derived from A and B. Define count () member function in all class as virtual. Count the number of objects created. (11)

UNIT IV

17. Write a C++ program to count the number of words and lines in a text file. (11)

Or

18. Explain Exception handling with an example program. (11)

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